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TRAINING REPORT

Duration- 12/05/2025 to 11/06/2025

Topic- SULPHURIC ACID PLANT(SAP)



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CERTIFICATE

This is to certify that the industrial training report has been submitted by **Srikant Pradhan**, Branch- **chemical engineering**, institute- **IGIT, SARANG**, under my guidance and supervision has completed the industrial training at IFFCO, Paradeep unit.

Submitted To:

Mr.K.P.Nanda Sir

Training Officer

IFFCO Paradeep

Submitted by:

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ACKNOWLEDGMENT

Any attempt at any level cannot be satisfactorily completed without the support and guidance of learned people. I take this opportunity to express my hearty thanks to all those who have individually or collectively helped me in the successful completion of this training report.

I sincerely thank IFFCO Paradeep for giving me the opportunity to undergo vocational training at their plant. It was a valuable learning experience that helped me understand real industrial processes.

I am grateful to **Mr. K.P. Nanda Sir** (Sec. Head Training), I/C Training and all the engineers and staff members for their guidance and support during the training period.

I would also like to thank the Chemical Engineering Department of IGIT, Sarang for encouraging practical learning through this internship.

Lastly, I thank my family and friends for their constant support.

PREFACE

In any organization success or failure of the company depend upon 4M's i.e, Materials, Men, Machine & Method. today is the age of competition & an organization can't survive without satisfaction of its customers. Quality of material is to be maintained in order to stand in the competitive market. To be a perfect engineer, one must be familiar with individual experience in industrial environment. He must be aware of basic industrial problems and remedies.

While undergoing this type of industrial training at FFCO, PARADEEP, Jagatsinghpur, Odisha. I learned a lot of practical aspect . My theoretical knowledge was exposed here practically. In this report one has tried to summarize what I have learned in the IFFCO plant. For preparing this report I visited the plant, referred to the process and cleared related doubts to the responsible personnel and inferred to manuals and process reports.

This study has been primarily undertaken by me with a view to evaluate proper working process in the organization.

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INTRODUCTION

The IFFCO Paradeep Plant, located in the industrial port town of Paradeep in Odisha, stands as a cornerstone of India's fertilizer production infrastructure. Established in 2005 through the strategic acquisition of Oswal Chemicals and Fertilizers Ltd., this facility marked the first instance of a cooperative sector entity acquiring a private sector asset in India . This move significantly bolstered IFFCO's capacity to produce complex fertilizers, aligning with the nation's growing agricultural demands.

Strategically situated on the eastern coast, Paradeep offers logistical advantages, including proximity to the Paradip Port, facilitating efficient import of raw materials and export of finished products. The plant's location also provides access to a skilled workforce and robust infrastructure, essential for large-scale industrial operations.

The Paradeep facility boasts impressive production capacities: 875,000 metric tons per annum (MTPA) of phosphoric acid, 2,310,000 MTPA of sulfuric acid, and 1,920,000 MTPA of diammonium phosphate (DAP) . Notably, it houses the world's largest single phosphoric acid reactor, with a daily production capacity of 2,650 tons of P_2O_5 . In the fiscal year 2021-22, the plant achieved a record production of 805,000 tons of phosphoric acid, the highest ever by a single plant globally .

Under the leadership of Shri P.K. Mahapatra, the plant has not only enhanced its production efficiency but also implemented significant environmental initiatives. In 2014, a heat recovery system was installed, generating an additional 65 tons per hour of steam, while efforts to reduce CO_2 emissions have led to a decrease of 131,372 tonnes per year .

Beyond its industrial prowess, the IFFCO Paradeep Plant plays a vital role in India's agricultural sector by ensuring the availability of high-quality fertilizers, thereby supporting the nation's food security and economic growth.

FIRE & SAFETY

Fire and Safety at IFFCO Paradeep Unit

During my internship at IFFCO Paradeep, I observed that fire and safety practices are given the utmost priority in all operations. As a large-scale fertilizer manufacturing unit dealing with hazardous chemicals like ammonia, phosphoric acid, and sulfuric acid, IFFCO maintains a robust safety management system to protect both personnel and plant assets.

1. Safety Management System

IFFCO Paradeep follows a comprehensive Safety, Health, and Environment (SHE) policy. All operations are carried out in compliance with the Factories Act, 1948, PESO (Petroleum and Explosives Safety Organisation) regulations, and OHSAS 18001 / ISO 45001 standards.

2. Fire Protection and Prevention

The plant is equipped with a highly responsive Fire and Safety Department that operates 24x7. Key fire safety installations include:

- Hydrant networks across all process and storage areas
- Fixed foam systems for tank farms
- CO₂ flooding systems in electrical panels
- Automatic fire alarm systems, smoke detectors, and heat sensors
- Portable fire extinguishers (CO₂, DCP, Foam, Water type) placed in accessible locations

Fire water storage tanks with diesel- and electric-driven fire pumps

3. Fire Station

IFFCO Paradeep has a dedicated fire station inside the plant premises equipped with:

- Fire tenders
- Ambulance services
- Trained firemen and rescue teams
- Breathing apparatus and chemical suits
- Routine inspections and equipment testing are done to ensure readiness

4. Mock Drills and Training

Regular mock drills (fire, toxic gas leak, explosion scenarios) are conducted in coordination with CISF (Central Industrial Security Force) to enhance emergency preparedness. All employees, including interns, receive induction training covering:

- Use of PPE (Personal Protective Equipment)
- Fire extinguisher handling
- Emergency response and assembly point protocols

5. Permit to Work (PTW) System

A strict Permit to Work system is followed for high-risk operations such as:

- Hot work (welding, cutting)
- Confined space entry
- Maintenance in flammable areas

This ensures proper risk assessment and supervisor approval before work begins.

SULPHURIC ACID PLANT(SAP)

IFFCO PARADEEP Sulfuric Acid plant (SAP) Is designed to produce **2 x 3500 MT**, 98.5% concentrated sulfuric acid per day. Manufacturing process is supplied by MIs Lurgi GmbH, Germany. The plant is operated on the basis of solid, bright sulfur intended for the production of H_2SO_4 .

Apart from the production of sulphuric acid, plant is designed to produce maximum amount of H.P superheated steam (60 Bar / 480°C) and L. P saturated steam (4 Bar / 150°C) with the utilization of the energy from the process.

The plant is designed to minimize the formation of nitrogen oxides during sulphur combustion (LURO), ie, achieve a high SO_2 conversion rate in a four layer converter, and observe the regulations of safety at work and environmental protection

Plant is controlled via DCS (Distributed Control System)

DCDA (Double Conversion Double Absorption) process is used in our Plant.

SAP DESIGN CRITERIA

Gas feed rate (1st bed inlet):	283436 NM ³ /Hr
SO₂ Concentration Upstream 1st bed:	11.5 (vol.%)
O₂ Concentration Upstream 1st bed :	9.23 (vol.%)
Overall Conversion from SO₂ to SO₃:	99.7 (vol.%)
Tail Gas SO₂ emission:	2.0 kg/MT of H ₂ SO ₄

PROCESS DESCRIPTION

Solid Sulphur is reclaimed from storage shed and transported to sulphur melting tanks after it is dozed with lime to neutralize the free acidity in it. The solid sulphur having purity of 99.5% on dry basis is melted in the tanks, which are provided with steam coil bundles, supplied with saturated steam at 6.0 kg/cm²(g). The liquid 'Sulphur' is filtered in 3 nos. of shell & leaf filters followed by filtration in candle type ceramic filtration unit. The filtered and pure liquid sulphur (>99.9% pure) is stored in the molten sulphur storage tank (MSST) at 1450 C. From this, it flows into pump tank from where required quantity of molten sulphur is pumped by individual pumps to the sulphur furnaces of two streams. The required air is filtered, dried in drying tower and supplied to the furnace by Turbine Driven Air Blower. Some part (about 5% to 7%) of air is supplied through Primary Air Fan, which atomizes the liquid sulphur for efficient burning. The sulphur is burnt in the Furnace at 11200C. The resulting SO₂ concentration is about 11.5% by volume. The corresponding O₂ content is about 9.2% by volume.



After the combustion, the gas is cooled down to 4200 C in a water tube forced circulation boiler, which is a Waste Heat Boiler where saturated steam at 61.2 kg/cm²(g) is generated. These gases pass

through the Converters I & II, which contain V₂O₅ catalyst. Each Converter has two beds of catalyst through which the SO₂ borne gases pass through and converted into SO₃.



The combustion gases at 4200C from Waste Heat Boiler will pass through the 1st bed of Converter-1. As SO₂ to SO₃ conversion is also exothermic; the gases pick up heat and come out at 6150 C with 59-60% SO₂ to SO₃ conversion achieved. These gases are cooled down to 4350 C in Super Heater-2 where partly super heated steam from Super Heater-1 is further super heated to 4800C at 61.2 kg / cm²(g). The cooling down of the gases between 4000C to 4500C is required prior to conversion as the SO₂ to SO₃ conversion reaction is equilibrium reaction dependent on the temperature. The gases at 4350C enter the Bed-2 of Converter-1 and come out at 5250C with an 83% of SO₂ to SO₃ conversion achieved. These hot gases are cooled down in Super Heater-1, where saturated steam from Waste Heat Boiler is partially super heated. The cooled outgoing gases at 4300C now enter Bed-3 of Converter-II, and come out at 4650 C with 95% SO₂ to SO₃ conversion achieved. Prior to entering into Bed-4 of Converter-II, the gases are cooled at 2160 C in Gas-Gas Heat Exchanger and enter into Inter Absorption Tower to Absorb SO₃ with recirculating 98.5%, H₂SO₄.

The humid air is dried prior to entering the sulphur combustion chamber and the contact group. This is done by trickling an approx. 96% sulphuric acid (54 deg. C) into a drying tower. This tower is a packed column with refractory lining where the air is injected from below in counter-current to the acid trickling down from above.

In this process, the sulphuric acid absorbs the water contained in the air and heats up to approx., 70°C depending on the temperature of the ambient air and its water concentration. The dried air leaves the tower through a wire-mesh filter installed in the “dome” to separate any acid droplets which may be contained in the gas.

The trickling acid from the air drying tower flows from the tower sump into the pump feed tank located on the same level. In this refractory lined tank, the concentration of the 96% H₂SO₄ Circuit is adjusted by adding an approx. 98.5% sulphuric acid from the intermediate and final absorber. Two vertical submersible acid pumps installed in the pump tank transfers the trickling acid for the drying tower through 4 (four) acid coolers (Plate-type heat exchanger) to the trickling system of the tower. Upstream of the cooler the level controlled excess drying acid is routed to the intermediate absorbing tower.

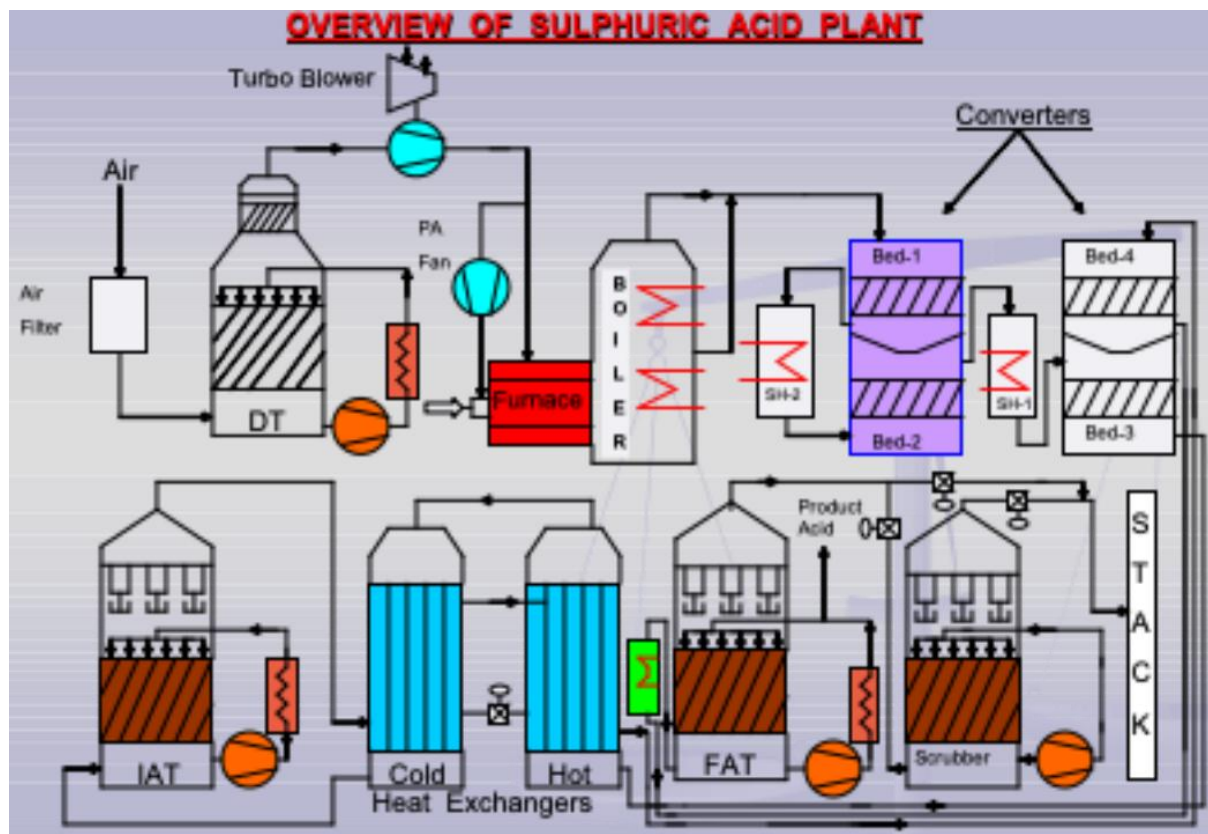
The part of the SO₃ generated in the converter layers 1 to 3 is absorbed in intermediate absorption tower to form 98.5%

sulphuric acid. This tower is a brick-lined packed tower operated at counter current with an approx. 98.5% sulphuric acid trickling. To adjust the concentration of the circulated sulphuric acid, 96% sulphuric acid from the air drying tower is added via control valve. Part of the product generated in the intermediate absorber unit is pumped to the final absorber as a function of the level in the pump feed tank where the product is withdrawn from the final absorber. Another part which mainly originates from the exchange volume from the dryer is returned into the drying tower via the level control of the intermediate absorption tower. Two pumps are installed in Pump Tank for acid circulation in Tower. Candle Filters are installed at the top of the trickling unit to separate acid droplets and mist from the gas downstream of the intermediate absorption tower.

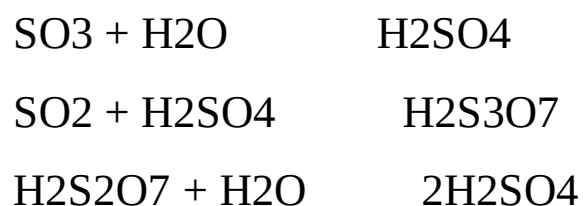
The SO₃ generated in the layer 4 of the Converter is absorbed in the Final absorption tower in an approx. 98.5% sulphuric acid. Final Absorption tower is connected with a pump tank in which two vertical submersible pumps are installed for acid circulation in tower and one pump is installed for taking out the product acid and sending it to the Storage Tank.

Each circulation pump discharge acid is routed through two plate type heat exchangers for cooling acid prior to trickling in acid tower. To adjust this concentration in the final absorption circuit,

96% sulphuric acid from air drying tower is added via a Control Valve.



At the top of final absorption tower trickling system candle filters are installed. The gas leaving the candle filter of the final absorption with acid mist concentration of 40 mg. SO₃ per NM₃ gas and SO₂ concentration of < 0.035% is routed through the tail gas stack to the atmosphere.



From each stream is generated 176.5 MTPH of superheated steam at 61.2kg/cm²(g) and 4800C.

The Plant is equipped with Alkali Scrubber Unit which is designed by FACT Engineering and Design Organization, Kerala. The objective of the scrubber unit is to treat the effluent gas during start up and to control the sulphur dioxide concentration within the prescribed limits. This system guaranties SO₂ concentration in outlet gas during plant start up to maximum of 200 mg / NM³.

PROCESS OF HEAT RECOVERY SYSTEM(HRS)

The Heat Recovery System (HRS) absorbs SO₃ from the gas stream and recovers the heat from the absorption process as intermediate pressure (IP) steam. The first stage recovers the heat of formation of sulfuric acid. The second stage removes the residual SO₃ and sulfuric acid vapor, and removes the remaining heat of formation, heat of condensation of sulfuric vapor, and sensible heat from cooling process gas. The inter pass tower serve as the 2nd stage of the HRS.

The HRS consists of a steam injection chamber, a packed heat recovery tower, a horizontal steam boiler, an HRS heater, an HRS pre-heater, an HRS diluter, an acid circulating pump (located in the pump boot), and acid drain pumps (2).

The steam injection chamber receives process gas from the existing acid plant. A portion of the water required to maintain HRS acid concentration is injected as low pressure steam (0.70 bar, minimum) into the process gas in the steam injection chamber. Steam injection has several advantages: decrease

corrosion rate in the first stage of the HRS tower and reduce acid flow over the first stage thereby reducing the size of the HRS diluter, HRS circulating pump and first stage distributor. Steam for injection will be let down from the outlet of the HRS boiler.

The heat recovery tower receives SO₃ gas from the steam injection chamber. SO₃ is absorbed from the gas flowing upward through the packing into circulated sulfuric acid flowing downward through the packing. The partially cooled gas, with the remaining SO₃, exits the HRS tower and enters the existing inter pass tower. The process gas flows upward through the existing inter pass tower packing, where essentially all of the remaining SO₃ is absorbed into circulated sulfuric acid flowing downward through the packing.

Acid from the first stage collects in the base of the tower and is circulated via a vertical, submerged, centrifugal pump installed in the pump boot.

Gas from the inter pass tower flows through mist eliminators installed in the top of the tower, and then to the heat exchanger, these mist eliminators remove acid mist formed in the towers down to a level equal to that of conventional inter pass absorbing tower mist eliminators, to protect the downstream heat exchanger.

Acid is pumped from the HRS tower pump boot to the horizontal boiler. From the boiler, most of the cooled acid flows to the HRS diluter and then back to the first stage of the HRS tower. A small stream, corresponding to the acid volume produced in the tower and any cross flow from the drying tower, is further cooled in the HRS heater and HRS pre-heater where heat is transferred to the HRS boiler feed water and treated water, respectively. From the HRS pre-heater, the acid flows to the existing acid system, where it is mixed with acid from the other towers.

Cooled acid from the acid cooler is supplied to the interpass tower at 98.5% minimum strength. Dilution water is continuously

mixed with acid downstream of the HRS boiler to control acid strength to a minimum of 99.0%. The acid is then circulated over the HRS tower first stage. The addition of dilution water and the resulting absorption of SO₃ raise the temperature of the circulated acid due to the heat of formation. As acid is circulated through the boiler, this heat is exchanged with boiling water. To provide a temperature driving force for heat exchange in the boiler, the acid is circulated at temperatures above the boiling point of the steaming water.

During periods of high humidity the amount of water vapor in the air entering the drying tower may be greater than the amount of H₂O required to control the final tower and product acid strength at 98.5%. At these times, increased flow of hot acid from the existing acid system will be sent to the HRS diluter. The HRS diluter is a Teflon lined stainless steel vessel.

The maximum recommended concentration for HRS circulated acid is 99.7%. At concentrations above this level, SO₃ absorption in the HRS tower decreases, resulting in reduced heat recovery. Because of the importance of concentration control, the HRS analyzers will utilize a PTFE-lined, flow through conductivity cell that is mounted externally on the sample lines. This eliminates the problem of probe breakage and erroneous readings from inadequate sample flow.

A comprehensive system of interlocks (actuated by acid temperature, strength, and flow) is provided to prevent continued operation with acid strength below the recommended minimum concentration of approximately 98.5% to the first stage.

The HRS boiler's continuous and intermittent blow down effluent is flashed to the atmosphere in the HRS boiler blow down flash tank and drain discharge is taken to the process sewer.

Raw Materials & Utilities

- Sulphur: Solid, Bright

Analysis On Dry Basis

Sulfur	Min 99.50 %wt
Ash	Max 0.05 %wt
Carbon	Max 0.05 %wt
H ₂ SO ₄	Max. 0.05 %wt
As	Nil
Se	Nil
Te	Nil
Hg	Nil
Cl-	Traces
H ₂ O Content	Max. 2.00 % wt
Size	Max. 100 mm

- **Boiler Feed Water (for HP Steam)**

Temperature (bat. Lim)	min.105	oC
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Pressure(bat. Lim)	min.91 bar g
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Analysis

Electrical conductivity (25 oC)	<0.2 μ S/cm
pH-value (25 oC)	9.5
O ₂	<0.02 mg/l
SiO ₂	<0.02 mg/l
Na ⁺	<0.01 mg/l
Fe (total)	<0.02 mg/l
Cu(total)	<0.003mg/l
Others	nil

Demineralized water

Temperature (bat. Lim.)	max.40oC
Pressure (bat. Lim.)	4 bar g

Analysis

Electrical conductivity	(25oC)	<0.3 μ S/cm
pH-value		7.0
O ₂		<0.1 mg/l
SiO ₂		<1.0 mg/l
CO ₂		<25.0 mg/l
Na ⁺		nil
Fe(total)		<0.05 mg/l
Cu(total)		<0.01 mg/l
Others		nil

Hardness	<0.03	m val/l
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Cooling water (cooling tower water)

Temperature in (bat. Lim)	max 35 oC
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Temperature out (bat.lim)	max 43 oC
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Pressure in (bat.lim)	min 5	bar g
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Pressure out (bat.lim)	min 2.5 bar g
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Analysis

Cl-	<200	mg/l
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Cl ₂	nil
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Organics	nil
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Raw water (Filtered water)

Temperature (bat.lim)	max 35 oC
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Pressure (bat.lim.)	min. 3 bar g
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Analysis

Dissolved solids	<132mg/l
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O ₂	<6 mg/l
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SiO ₂	<14 mg/l
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Cl-	<14 mg/l
-----	----------

CaO	<19 mg/l
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HP steam for blower turbine start up-

Superheated steam

Pressure (bat.lim.)	min.	60	bar g
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Temperature (bat.lim.)	400 oC
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HP/LP steam for preheating purposes of converter beds

Slightly superheated steam

Pressure 8 to 40 bar g

Temperature 255 oC

LP steam for sulphur melting

Saturated steam min. 5 bar g

Pressure Max. 6.5 bar g

LP steam for liquid sulphur heating

Saturated steam min. 3.5 bar g

Pressure Max. 4.0 bar g

Instrument air

Temperature max 35 oC

Pressure min 5 bar g

Water dew point <-40 oC

Oil content nil

Fuel oil for plant preheating

Type: HSD

Temperature (bat.lim.) Max.35 oC

Pressure (bat.lim.) 2 bar g

Heat of combustion min.41.9 MJ/kg

Analysis

Ash **<0.01 %wt**

Sulphur **<0.2 %wt**

Igniton gas

Type: LPG

Lime for neutralization of sulphur

Analysis

Ca (OH)₂ min. 98.0 %wt

Filter aid

Filter aid for sulphur filtration qualities are depending on the quality of the sulphur which has to be filtered and on the wanted quality of the filtered sulphur.

Usual are the following types/qualities of filter aid (diatomaceous earth):

Hyflo Supercel, Celite 503 or Celite 535

Speedplus, speedex or dicalite 4200

Clarcel Dic-s, Clarcel DIF, Clarcel DIT-II-R

Sulfur Melting, Filtration and storage

The sulphur melting area is organized and equipped to permit the safe handling of solid and molten materials to provide the necessary clean feedstock of molten sulphur to the sulphuric acid plant.

Solid Sulfur Feed

Solid sulphur is fed continuously into the melting tank via feed hoppers, vibrofeeder and belt conveyor. A central agitator installed in the sulphur melter recirculates liquid sulphur through the heating coils. The solid sulphur is immersed into sulphur stream and

transported to the heating coils thus ensuring a prompt and efficient melting of the solid sulfur.

Sulfur Melting

The solid sulphur is transferred to the feed hoppers by means of a front-end loader. Using belt conveyors, solid sulphur is sent to the melting tanks (010V0130/131/132). The sulphur is melted by means of steam-heated tube bundles. The agitators (010A0130/131/132) are arranged in the centre provides for uniform distribution of the sulphur in the tanks. Important points in sulphur operation

Inadequate heating of the sulphur piping and valves may lead to clogging as a result of solidified sulphur. It is therefore essential to see the functioning of the steam traps installed. It is important that the sulphur temperature is at all times kept between 130-150°C. This depends on the pressure and temperature of the heating steam. Saturated steam at 3.5 Kg/cm²(g) and temperature 145 °C is used. Sulphur temperature is never allowed to go above 155 °C because in this range the sulphur becomes highly viscous and may lead to pumping problem.

Sulphur filter (010S0140/141/142)

Ash content in molten sulfur may cause severe problem in down stream process, like they may cause pipe choking, deposit on catalyst bed that reduces the conversion efficiency, so it is necessary to remove ash from the molten sulfur. To remove the ash content from molten sulfur, it is passed through primary and secondary filter.

Major consequences of poor filtration are as follow:-

Premature fouling of catalyst bed

Corrosion on down stream line

High viscosity effects on rate and efficiency

By means of the pumps (010P0136/137/236/237/336/337) installed in the pump tanks (010V0136/236/336), sulphur is extracted from this tank by pumping it through the sulphur filter where it is filtered and

the pure sulphur is transferred to the storage tank for intermediate storage.

In order to be able to achieve optimum filtration results, it is necessary to abide by the following criteria:

Filter coating correctly applied.

Make sure that the filters are continuously fed with sulphur; if the sulphur feed is interrupted, it is necessary to empty the filters and apply again a filter coat; when the sulphur feed is restarted in this case, unfiltered sulphur may leak into “pure sulphur tank”.

At a pressure in the filter of about 5.5 bar (g), stop filtration, empty the filter, clean it and precoat it again .

Make sure that the filters and the pertinent equipment are always kept well heated.

In principle, a filtering time of about 16 h/d has been planned as well as 8 h/d for cleaning, Precoating, etc.

Filtered Sulfur Storage

The pure sulphur coming from the filtering station is stored in Molten Sulfur Storage Tank (010T0160/161). From the storage tank, the liquid sulphur is transferred to a pump tank from which it is routed to the sulphur burner by means of vertical submersible pumps.

In order to ensure proper operation, following is to be taken in account:

The storage tank is provided with level indicator. Under normal conditions the melting and the filtration are to be controlled such that the level in the tank is maintained in the upper third; in order to avoid overfilling, it is necessary to reduce the capacity or disconnect the sulphur melting/filtration units.

By means of installed heating coils, the sulphur temperature in the storage tank is maintained at about 150°C; it is necessary to ensure that the steam valves in the inlet piping are completely open; the proper function of the condensate separators has to be ensured.

The storage tank is connected with a pump tank (010V0160/260). The filling level in the pump tank is maintained constant by adding sulphur from the storage tank by means of control valve (LUCAS JL0105/0205); opening the valve means a higher level; closing the valve means a lower level.

From the pump tank, the sulphur is sent to the sulphur burner. It is to be ensured that there is sufficient heating for the pump tank as well as the piping including the circulation line.

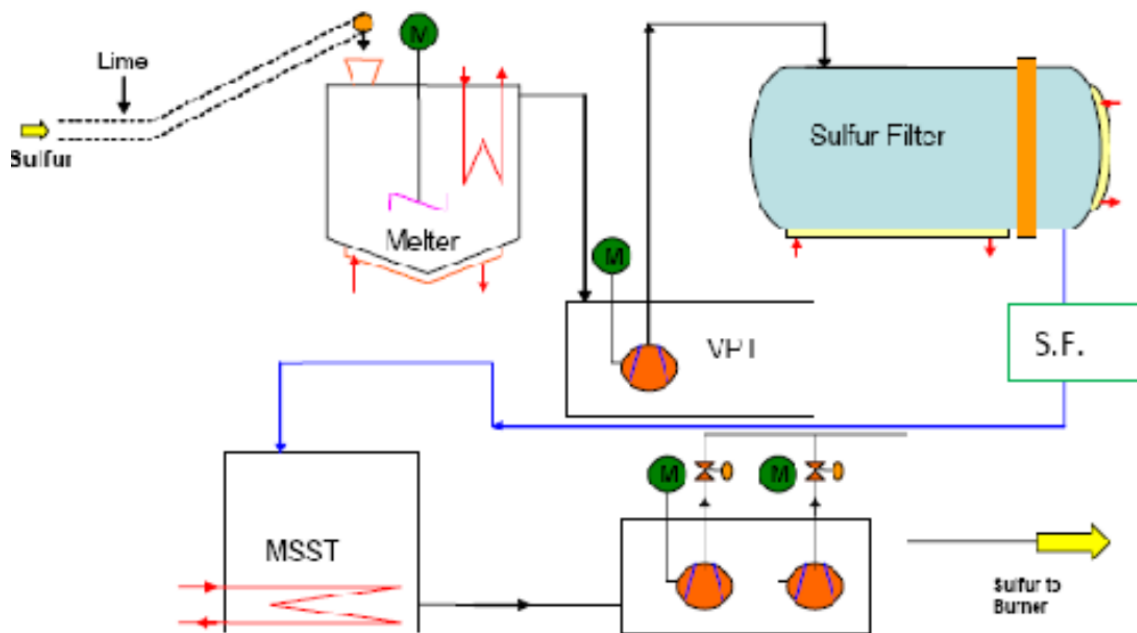
Start up of the sulphur melting, filtration and storage facilities

All instruments should be in operation.

Before commencing the starting procedures for each individual system of the sulphuric acid plant, it should be ensured that this particular unit is mechanically complete, which in general means the following:

- Verify that all sulphur pits and the storage tank are clean and dry
- Disconnect all steam blocks in relation to the piping and heating coils
- Purge all steam and condensate lines with steam
- As soon as these measures have been taken, the steam blocks can again be connected. The steam and condensate systems in connection with the sulphur system will then be ready for testing and preheating, as follows:
- Start the various steam systems by opening the respective valves successively
- Check every system for possible leaks. Piping with steam jackets should be inspected to detect internal leaks
- Verify every steam lock to check if it works properly
- After heating, it is essential to check all flanged connections at piping and equipment, retightening the bolts if necessary
- As soon as the sulphur system has been correctly reheated, it may be charged with sulphur.

MELTING AREA



Sulphur melting

If the heating system for all heating systems (heating coil of the melting tanks and of the storage tank, double-jacket piping, valves, fittings, etc.) has been in operation, sulphur melting can be started.

- Fill feed hoppers with sulphur, about 50%
- Fill lime hoppers with lime

Start belt conveyors for sulphur after blowing hooter.

Check belt interlock and pull cord.

- Start electromagnet installed at transfer points.
- Start lime dosing; adjust the lime rate as a function of the acid content of the sulphur.

Start agitators after ensuring its freeness once molten sulphur level in melter reaches 50 %.

- Sulfur filtration
- Filing a filter

One of the sulphur pumps installed in the sulphur pump tank (010V0136/236/336) is used to fill the filter. The following valves should be totally open for filling the filter:

- For pressure indication in sulphur filter (PIA JP104/6)
- In the piping upstream of filter (010S0140/2)
- In the filter ventilation ducts
- In the recirculation line

Completely closed:

- In the piping towards the storage tank
- In the drain line of filter.

Preparations for applying the filter coat to the filter plates.

Close the sulphur valves completely

At the ventilation

Start the agitator (010A0136/236/336) in the pump tank and check its proper functioning.

Put about 20 kg of filter aid (diatomaceous earth) and hydrated lime powder (as per requirement by seeing sulphur acidity) into the precoating tank.

The agitator will mix the sulphur and the powder within a few minutes.

Check the pump operation. While the sulphur circulates from the pump tank through piping towards the sulphur filter and returns to the pump tank through piping, the balance of 60 kg diatomaceous earth should be slowly added to the pump tank. The coating time is about 45 minutes.

As soon as the sulphur circulating in the tank has changed its red colour to dark brown, the total filter aid and lime remain deposited on the filter plates.

Start of filtration

Open the valve in piping towards the storage tank (010T0160) slowly until it is half open and slowly close recirculation line valve. For

sulphur melting and filtration, the operating time should be approximately 16 hours. The maximum allowable pressure in the sulphur filter is about 5.5 bars.

Filling the sulphur storage tank (010T0160/161) and the pump tank (010V0160/260)

The storage tank (010T0160) is filled with filtered sulphur by means of pipeline from sulphur filter discharge.

Sulphur burning should start after the tank is filled with sulphur.

If the level in this tank is too high, it is essential to reduce or stop the feed to the melting tank and subsequently also the filter.

Pump tank 010V0160/260 is linked with the storage tank. For filling the pump tank valves being opened completely and the entering rate is controlled by control valve connected to (LUCAS JL0105/0205).

As soon as the level has reached 60%, the filling operation is stopped.

Pump 010P0160/161/260/261 is started and the corresponding valve after the pump opened.

Attention:

Pumps 010P0160/161/260/261 in normal operation must be interlocked with the air blower; for this purpose, the interlock changeover switch of the pump should be set at “INTERLOCKED” position after starting the plant.

DESIGN DATA

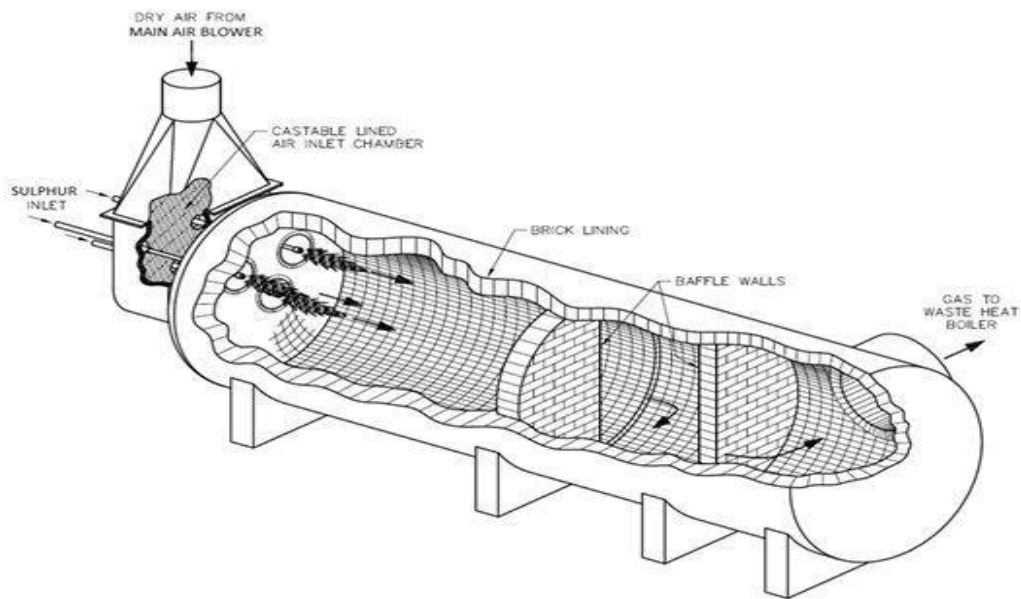
Capacity	2×3500 MTPD	2000 MTPD
Technology	M/S LURGI GMBH (Germany)	MECS(USA)
Consultant	H&G	SSS (Shiv Sulphuric Solution)
Gas feed rate (bed-1)	283436 NM ³ /Hr	173027 NM ³ /Hr
SO ₂ concentration upstream	11.5 (vol.%)	10.7 (vol.%)
O ₂ concentration upstream	9.23 (vol.%)	9.83 (vol.%)
SO ₂ conversion	99.7 (vol.%)	99.92 (vol.%)
Tail gas SO ₂ concentration	0.03 (vol.% max)	0.01 (vol.% max)
Allowable SO ₂ concentration	2 KG/Ton of H ₂ SO ₄	0.5 KG/Ton of H ₂ SO ₄
Allowable SO ₂ emission at stack	400 ppm	100 ppm

SAP-3 FURNACE

In Tr-3 internal baffles are provided for mixing which leads to proper gas combustion whereas in Tr-1 & 2 baffles are not required because already atomization happens with help of PA fan and rotary cup burner followed by desired combustion.

Length of furnace is more for maximum residence time in Tr-3 as compared to furnace of Tr-1 & 2.

Five numbers of ports are required for guns in SAP r-3 whereas only two ports are sufficient in tr1&2.



TR-3 FURNACE 3D VIEW



TR-3 FURNACE INSIDE VIEW

EXPERIENCE:

We are thankful to engineer on the site under whose valuable guidance we were able to get a view of our theoretical knowledge being converted into a practical one.

CONCLUSION:

Being not only India's but also Asia's premiere fertilizer producing company, the Paradeep unit of IFFCO achieved the highest ever yearly sulphuric acid plant capacity utilisation of 82.25% surpassing the previous highest of 61.83% achieved in 2009-10.

It was an absolute privilege and honour working for such a cooperation. Their mission is "to enable Indian farmers to prosper through timely supply of reliable, high quality agricultural inputs and services in an environmentally sustainable manner and to undertake other activities to improve their welfare".

As is their mission so noble so is their dedication towards their work. The experience we gained here is very valuable. And I am thankful to each and everyone involved to make this training a successful one.

THANKYOU!!

